KAIZEN MATHS

Lesson starter

Friction and Rough Surfaces

Rough Horizontal Motion

Name _____

S1

A block of mass 12 kg is pulled horizontally by a force of 100 N on a rough horizontal surface. The coefficient of friction is 0.25. Find the acceleration. Take g = 9.8 m s^{-2}.

S2

A block of mass 12 kg accelerates at 0.5 m s^{-2} on a rough horizontal surface. The coefficient of friction is 0.25. Find the horizontal pulling force. Take g = 9.8 m s^{-2}.

S3

A block is moving at 8 m s^{-1} on a rough horizontal surface. The coefficient of friction is 0.25. Find the stopping distance. Take g = 9.8 m s^{-2}.

S4

A block of mass 15 kg is pulled by a force of 80 N at 20° above the horizontal. The coefficient of friction is 0.15. Find the acceleration. Take g = 9.8 m s^{-2}.

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A block of mass 12 kg is pulled horizontally by a force of 100 N on a rough horizontal surface. The coefficient of friction is 0.25. Find the acceleration. Take g = 9.8 m s^{-2}.

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A block of mass 12 kg accelerates at 0.5 m s^{-2} on a rough horizontal surface. The coefficient of friction is 0.25. Find the horizontal pulling force. Take g = 9.8 m s^{-2}.

S3

A block is moving at 8 m s^{-1} on a rough horizontal surface. The coefficient of friction is 0.25. Find the stopping distance. Take g = 9.8 m s^{-2}.

S4

A block of mass 15 kg is pulled by a force of 80 N at 20° above the horizontal. The coefficient of friction is 0.15. Find the acceleration. Take g = 9.8 m s^{-2}.

MEC-DED4-L2 · S

Show your working

KAIZEN MATHS

Exit ticket

Friction and Rough Surfaces

Rough Horizontal Motion

Name _____

EA1

A block of mass 15 kg is pulled horizontally by a force of 120 N on a rough horizontal surface. The coefficient of friction is 0.3. Find the acceleration. Take g = 9.8 m s^{-2}.

EA2

A block of mass 20 kg accelerates at 1.5 m s^{-2} on a rough horizontal surface. The coefficient of friction is 0.3. Find the horizontal pulling force. Take g = 9.8 m s^{-2}.

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KAIZEN MATHS

Exit ticket

Friction and Rough Surfaces

Rough Horizontal Motion

Name _____

EB1

A block is moving at 10 m s^{-1} on a rough horizontal surface. The coefficient of friction is 0.25. Find the stopping distance. Take g = 9.8 m s^{-2}.

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KAIZEN MATHS

Exit ticket

Friction and Rough Surfaces

Rough Horizontal Motion

Name _____

EA1

A block of mass 15 kg is pulled horizontally by a force of 120 N on a rough horizontal surface. The coefficient of friction is 0.3. Find the acceleration. Take g = 9.8 m s^{-2}.

EA2

A block of mass 20 kg accelerates at 1.5 m s^{-2} on a rough horizontal surface. The coefficient of friction is 0.3. Find the horizontal pulling force. Take g = 9.8 m s^{-2}.

MEC-DED4-L2 · EA

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KAIZEN MATHS

Exit ticket

Friction and Rough Surfaces

Rough Horizontal Motion

Name _____

EB1

A block is moving at 10 m s^{-1} on a rough horizontal surface. The coefficient of friction is 0.25. Find the stopping distance. Take g = 9.8 m s^{-2}.

MEC-DED4-L2 · EB

Show your working

Teacher answers & working

Friction and Rough Surfaces · Rough Horizontal Motion

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S1

A block of mass 12 kg is pulled horizontally by a force of 100 N on a rough horizontal surface. The coefficient of friction is 0.25. Find the acceleration. Take g = 9.8 m s^{-2}.

Answer:

5.883 m s^{-2}

Working:

Find friction.

$R = mg = 12 \times 9.8 = 117.6$ N

$F = \mu R = 0.25 \times 117.6 = 29.4$ N

Apply Newton's second law horizontally.

Resultant force = 100 − 29.4 = 70.6 N

$ma = 70.6$

$12a = 70.6$

$a = \frac{70.6}{12}$

$a = 5.883$ m s^{-2}

S2

A block of mass 12 kg accelerates at 0.5 m s^{-2} on a rough horizontal surface. The coefficient of friction is 0.25. Find the horizontal pulling force. Take g = 9.8 m s^{-2}.

Answer:

35.4 N

Working:

Find friction.

$R = mg = 12 \times 9.8 = 117.6$ N

$F = \mu R = 0.25 \times 117.6 = 29.4$ N

Pulling force must overcome friction and produce acceleration.

$P - F = ma$

$P - 29.4 = 12 \times 0.5$

$P = 6 + 29.4$

$P = 35.4$ N

Teacher answers & working

Friction and Rough Surfaces · Rough Horizontal Motion

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S3

A block is moving at 8 m s^{-1} on a rough horizontal surface. The coefficient of friction is 0.25. Find the stopping distance. Take g = 9.8 m s^{-2}.

Answer:

13.061 m

Working:

Work done against friction equals the initial kinetic energy. The mass cancels.

Friction = μmg

$\mu mgs = \frac{1}{2} m\ u^2$

Cancel m: μgs = $\frac{1}{2}u^2$

$s = \frac{u^2}{2\mu g}$

$s = \frac{8^2}{2 \times 0.25 \times 9.8}$

$s = 13.061$ m

S4

A block of mass 15 kg is pulled by a force of 80 N at 20° above the horizontal. The coefficient of friction is 0.15. Find the acceleration. Take g = 9.8 m s^{-2}.

Answer:

3.815 m s^{-2}

Working:

Resolve vertically to find the normal reaction.

$R + 80 \sin 20° = mg$

$R = 15 \times 9.8 - 80 \sin 20°$

$R = 119.638$ N

Find friction and the horizontal component of the pull.

$F = \mu R = 0.15 \times 119.638 = 17.946$ N

Horizontal pull = 80 cos 20° = 75.175 N

Apply Newton's second law horizontally.

$75.175 - 17.946 = 15a$

$a = \frac{57.23}{15}$

$a = 3.815$ m s^{-2}

Teacher answers & working

Friction and Rough Surfaces · Rough Horizontal Motion

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EA1

A block of mass 15 kg is pulled horizontally by a force of 120 N on a rough horizontal surface. The coefficient of friction is 0.3. Find the acceleration. Take g = 9.8 m s^{-2}.

Answer:

5.06 m s^{-2}

Working:

Find friction.

$R = mg = 15 \times 9.8 = 147$ N

$F = \mu R = 0.3 \times 147 = 44.1$ N

Apply Newton's second law horizontally.

Resultant force = 120 − 44.1 = 75.9 N

$ma = 75.9$

$15a = 75.9$

$a = \frac{75.9}{15}$

$a = 5.06$ m s^{-2}

EA2

A block of mass 20 kg accelerates at 1.5 m s^{-2} on a rough horizontal surface. The coefficient of friction is 0.3. Find the horizontal pulling force. Take g = 9.8 m s^{-2}.

Answer:

88.8 N

Working:

Find friction.

$R = mg = 20 \times 9.8 = 196$ N

$F = \mu R = 0.3 \times 196 = 58.8$ N

Pulling force must overcome friction and produce acceleration.

$P - F = ma$

$P - 58.8 = 20 \times 1.5$

$P = 30 + 58.8$

$P = 88.8$ N

Teacher answers & working

Friction and Rough Surfaces · Rough Horizontal Motion

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EB1

A block is moving at 10 m s^{-1} on a rough horizontal surface. The coefficient of friction is 0.25. Find the stopping distance. Take g = 9.8 m s^{-2}.

Answer:

20.408 m

Working:

Work done against friction equals the initial kinetic energy. The mass cancels.

Friction = μmg

$\mu mgs = \frac{1}{2} m\ u^2$

Cancel m: μgs = $\frac{1}{2}u^2$

$s = \frac{u^2}{2\mu g}$

$s = \frac{10^2}{2 \times 0.25 \times 9.8}$

$s = 20.408$ m